Solution

Stratum	Homogenous group	Percentage from population	Number of people in each strata	Random Samples
Category1	Lower income class	39	$\frac{39}{100}\times 500 = 195$	$195\times\frac{68}{500} = 26.5 \sim 26$
Category2	Middle income class	38	$\frac{38}{100}\times 500 = 190$	$190\times\frac{68}{500} = 26.5 \sim 26$
Category3	Upper income class	23	$\frac{23}{100}\times 500 = 115$	$115\times\frac{68}{500} = 15.6 \sim 16$
Total		**100**	**500**	**68**

Merits

(a) A random stratified sample is superior to a simple random sample because it ensures representation of all groups and thus it is more representative of the population which is being sampled.

(b) A stratified random sample can be kept small in size without losing its accuracy.

(c) It is easy to administer,if the population under study is sub-divided.

(d) It reduces the time and expenses in dividing the strata into geographical divisions, since the government itself had divided the geographical areas.

Demerits

(a) To divide the population into homogeneous strata (if not divided), it requires more money, time and statistical experience which is a difficult one.

(b) If proper stratification is not done, the sample will have an effect of bias.

(c) There is always a possibility of faulty classification of strata and hence increases variability.

(iii) Systematic Sampling

Definition 8.2

In a systematic sampling, randomly select the first sample from the first k units. Then every k^{th} member, starting with the first selected sample, is included in the sample.

Systematic sampling is a commonly used technique, if the complete and up-to-date list of the sampling units is available. We can arrange the items in numerical, alphabetical, geographical or in any other order. The procedure of selecting the samples starts with selecting the first sample at random, the rest being automatically selected according to some pre-determined pattern. A systematic sample is formed by selecting every item from the population, where k refers to the sample interval. The sampling interval can be determined by dividing the size of the population by the size of the sample to be chosen. That is $k = \frac{N}{n}$, where k is an integer.

k = Sampling interval, N = Size of the population, n = Sample size.

Procedure for selection of samples by systematic sampling method

(i) If we want to select a sample of 10 students from a class of 100 students, the sampling interval is calculated as $k = \frac{N}{n} = \frac{100}{10} = 10$.

Thus sampling interval = 10 denotes that for every 10 samples one sample has to be selected.

(ii) The firstsample is selected from the first 10 (sampling interval) samples through random selection procedures.

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